www.Padasalai.Net www.TrbTnpsc.com

P

COMMON QUARTERLY EXAMINATION - 2025

Standard - 12

Reg.No. | | | | | | |

MATHEMATICS

Time Allowed:3.00 Hours *Maximum Marks:90*

PART-A

i) Answer all the questions. **20×1=20**

ii) Choose the most suitable answer from the given four alternatives.

1) If $A = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ x & \frac{3}{5} \end{bmatrix}$ and $A^T = A^{-1}$, then the value of x is

a) $\frac{-4}{5}$ b) $\frac{-3}{5}$ c) $\frac{3}{5}$ d) $\frac{4}{5}$

2) If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and $A(\text{adj } A) = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$, then k =

a) 0 b) sin θ c) cos θ d) 1

3) If $\rho(A) < \rho([A/B])$, and then the system AX = B of linear equations is

a) consistent and has a unique solution

b) consistent

c) consistent and has infinitely many solutions

d) inconsistent

4) If $|z| = 1$, then the value of $\frac{1+z}{1+\bar{z}}$ is

a) z b) $\bar{z}$ c) $\frac{1}{z}$ d) 1

5) If $\frac{z-1}{z+1}$ is purely imaginary, then |z| is

a) $\frac{1}{2}$ b) 1 c) 2 d) 3

6) If α and β are roots of $x^2 + x + 1 = 0$, then $\alpha^{2020} + \beta^{2020}$ is

a) −2 b) −1 c) 1 d) 2

7) If f and g are polynomials of degrees m and n respectively, and if h(x) = (f o g)(x), then the degree of h is

a) mn b) m + n c) m^n d) n^m

8) If $x^3 + 12x^2 + 10ax + 1999$ definitely has a positive zero, if and only if

a) $a \geq 0$ b) $a > 0$ c) $a < 0$ d) $a \leq 0$

9) The polynomial equation $4x^4 - 3x^3 + 3x - 4 = 0$ is

a) odd degree reciprocal equation of Type II

b) even degree reciprocal equation of Type II

c) even degree reciprocal equation of Type I

d) odd degree reciprocal equation of Type I

10) If $\cot^{-1} x = \frac{2\pi}{5}$ for some $x \in R$, then the value of $\tan^{-1} x$ is

a) $-\frac{\pi}{10}$ b) $\frac{\pi}{5}$ c) $\frac{\pi}{10}$ d) $-\frac{\pi}{5}$

Kindly Send Me Questions & Answer Keys to Us: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

11) If $\cot^{-1} 2$ and $\cot^{-1} 3$ are two angles of a triangle, then the third angle is

a) $\frac{\pi}{4}$ b) $\frac{3\pi}{4}$ c) $\frac{\pi}{6}$ d) $\frac{\pi}{3}$

12) If $\sin^{-1}\frac{x}{5} + \text{cosec}^{-1}\frac{5}{4} = \frac{\pi}{2}$, then the value of x is

a) 4 b) 5 c) 2 d) 3

13) The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

14) The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is

a) $\frac{4}{3}$ b) $\frac{4}{\sqrt{3}}$ c) $\frac{2}{\sqrt{3}}$ d) $\frac{3}{2}$

15) If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is

a) 3 b) −1 c) 1 d) 9

16) The locus of a point whose distance from (−2,0) is $\frac{2}{3}$ times its distance from the line $x = \frac{-9}{2}$ is

a) a parabola b) a hyperbola
c) an ellipse d) a circle

17) If $\vec{a}$ and $\vec{b}$ are unit vectors such that $\left[\vec{a}, \vec{b}, \vec{a} \times \vec{b}\right] = \frac{1}{4}$, then the angle between $\vec{a}$ and $\vec{b}$ is

a) $\frac{\pi}{6}$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{3}$ d) $\frac{\pi}{2}$

18) The angle between the lines $\frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2}$ is

a) $\frac{\pi}{6}$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{3}$ d) $\frac{\pi}{2}$

19) If the plane $\vec{r}.(2\hat{i} - \lambda\hat{j} + \hat{k}) = 3$ and $\vec{r}.(4\hat{i} + \hat{j} - \mu\hat{k}) = 5$ are parallel, then the value of λ and μ is

a) $\frac{1}{2}, -2$ b) $-\frac{1}{2}, 2$ c) $-\frac{1}{2}, -2$ d) $\frac{1}{2}, 2$

20) If $\vec{a}$ and $\vec{b}$ are parallel vectors, then which one of the following are true?

i) $\vec{a} \cdot \vec{b} = 0$ ii) $\vec{a} \times \vec{b} = 0$
iii) $\vec{a} = \lambda\vec{b}$ iv) $\vec{b} = \lambda\vec{a}$

a) (i) and (iii) b) (ii) and (iii)
c) only (ii) d) (ii), (iii) and (iv)

Kindly Send Me Questions & Answer Keys to Us: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

PART-B

i) Answer any seven questions. **7×2=14**

ii) Question No.30 is compulsory.

21) Solve the following systems of linear equations by Cramer's rule : $5x - 2y + 16 = 0$, $x + 3y - 7 = 0$.

22) If A is non-singular matrix of odd order, prove that |adj A| is positive.

23) Obtain the cartesian equation for the locus of z in which $|z| = |z - i|$.

24) Find the square roots of $4 + 3i$.

25) If α, β and γ are the roots of the cubic equation $x^3 + px^2 + qx + r = 0$, find the value of $\sum \frac{1}{\beta\gamma}$ in terms of the coefficients.

26) Find centre and radius of the following circles. $2x^2 + 2y^2 - 6x + 4y + 2 = 0$.

27) Find the equation of the parabola whose vertex is (5,−2) and focus (2,−2).

28) Find the principal value of $\sec^{-1}(-\sqrt{2})$.

29) Find the volume of the parallelopiped whose coterminous edges are represented by the vectors $-6\hat{i}+14\hat{j}+10\hat{k}$, $14\hat{i}-10\hat{j}-6\hat{k}$ and $2\hat{i}+4\hat{j}-2\hat{k}$.

30) If $\tan^{-1} x + \tan^{-1} y = \pi$, show that $x + y = 0$.

PART-C

i) Answer any seven questions. **7×3=21**

ii) Question No.40 is compulsory. $-\sqrt{2}$

31) If $A = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix}$ verify that $(AB)^{-1} = B^{-1}A^{-1}$.

32) Four men and 4 women can finish a piece of work jointly in 3 days while 2 men and 5 women can finish the same work jointly 4 days. Find the time taken by one man alone and that of one women alone to finish the same work by using matrix inversion method.

33) Find the value of the real numbers x and y, if the complex number $(2 + i)x + (1 - i)y + 2i - 3$ and $x + (-1 + 2i)y + 1 + i$ are equal.

34) State and prove triangle inequality.

35) It is known that the roots of the equation $x^3 - 6x^2 - 4x + 24 = 0$ are arithmetic progression. Find its roots.

36) Find the value of $\cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right) + \cos^{-1}\left(\cos\left(\frac{5\pi}{4}\right)\right)$.

37) Find the equation of the ellipse with foci (±3,0) and $e = \frac{1}{2}$.

38) Find the points where the straight line passes through (6,7,4) and (8,4,9) cuts the xz and yz planes.

39) Find the angle between the planes $\vec{r}.\left(\hat{i}+\hat{j}-2\hat{k}\right) = 3$ and $2x - 2y + z = 2$.

40) Prove that the length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{2b^2}{a}$.

Kindly Send Me Questions & Answer Keys to Us: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

PART-D

i) Answer all the questions: **7×5=35**

41) a) Investigate for what values of λ and μ the system of linear equations has $x + 2y + z = 7$, $x + y + \lambda z = \mu$, $x + 3y - 5z = 5$
i) no solution ii) a unique solution
iii) an infinite number of solutions **(OR)**

b) Find the value of $\sin\left(\tan^{-1}\left(\frac{1}{2}\right) - \cos^{-1}\left(\frac{4}{5}\right)\right)$.

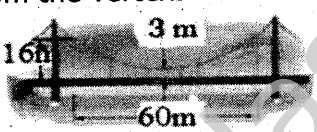
42) a) Find all cube roots of $\sqrt{3}+i$. **(OR)**

b) If $ax^2 + bx + c$ is divided by $x + 3$, $x - 5$ and $x - 1$, the remainders are 21, 61 and 9 respectively. Find a, b and c. (use Gaussian elimination method)

43) a) If $2 + i$ and $3 - \sqrt{2}$ are roots of the equation $x^6 - 13x^5 + 62x^4 - 126x^3 + 65x^2 + 127x - 140 = 0$, find all roots. **(OR)**

b) Show that the line $x - y + 4 = 0$ is a tangent to the ellipse $x^2 + 3y^2 = 12$. Also find the coordinates of the point of contact.

44) a) Parabolic cable of a 60 m portion of the roadbed of a suspension bridge are positioned as shown below. Vertical cables are to be spaced every 6 m along this portion of the roadbed. Calculate the length of first vertical cables from the vertex.



(OR)

b) Show that the lines $\frac{x-3}{3} = \frac{y-3}{-1}$, $z - 1 = 0$ and $\frac{x-6}{2} = \frac{z-1}{3}$, $y - 2 = 0$ intersect. Also find the point of intersection.

45) a) If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$ and $0 < x, y, z < 1$, then show that $x^2 + y^2 + z^2 + 2xyz = 1$. **(OR)**

b) A tunnel through a mountain for a fourlane highway is to have a elliptical opening. The total width of the highway (not the opening) is to be 16 m, and the height at the edge of the road must be sufficient for a truck 4 m high to clear if the highest point of the opening is to be 5 m approximately. How wide must the opening be?

46) a) If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{j} + 2\hat{k}$, $\vec{c} = -\hat{i} - 2\hat{j} + 3\hat{k}$ verify that $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a}.\vec{c})\vec{b} - (\vec{a}.\vec{b})\vec{c}$. **(OR)**

b) Find the non-parametric form of vector equation and cartesian equation of the plane passing through the point (1, −2, 4) and perpendicular to the plane $x + 2y - 3y = 11$ and parallel to the straight line $\frac{x+7}{3} = \frac{y+3}{-1} = \frac{z}{1}$.

47) a) Find centre, vertices and foci of the hyperbola $13x^2 - 49y^2 + 182x + 98y - 49 = 0$. **(OR)**

b) If $z = x + iy$ is a complex number such that $\text{Re}\left(\frac{z-i}{z+2}\right) = 2$, show that the locus of z is $x^2 + y^2 + 6x + y + 8 = 0$.

- - - - -

Kindly Send Me Questions & Answer Keys to Us: padasalai.net@gmail.com